Security Operations Center (SOC) Alert Investigation using SIEM – Simulation Report

1. **Introduction**

- Brief about cyber security monitoring in organizations (like banks)
- Role of a SOC analyst
- Purpose of using a SIEM tool

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2. **Scenario Overview**

- Description of the bank environment
- Role assigned: SOC Analyst
- Objective: Identify malicious vs non-malicious alerts

3. Understanding SIEM

- What is a SIEM (Security Information and Event Management)
- How SIEM collects logs from multiple sources
- Alert generation process

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4. **Types of Alerts Observed**

- Failed login attempts
- Suspicious IP connections
- Other abnormal activities

5. Alert Investigation Process

- Step-by-step analysis approach
- Checking logs and patterns
- Identifying false positives vs real threats

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6. SIEM Simulation Activity (Try Hack Me)

- Overview of the simulation
- Steps followed in the platform
- Key findings during investigation

Not all alerts are malicious, however. It is up to the analyst to use their expertise in cyber security to investigate which ones are harmful.

For example, you may encounter an alert where a user has failed multiple login attempts. While suspicious, this kind of thing happens, especially if the user has forgotten their password and continues to try to log in.

Additionally, there might be alerts related to connections from unknown IP addresses. An IP address is like a home address for your computer on the Internet—it tells other computers where to send the information you request. When these addresses are unknown, it could mean that someone new is trying to connect or someone is attempting unauthorized access.

Simulating a SIEM

We have prepared a simplified, interactive simulation of a SIEM system to provide you with a hands-on experience similar to what cyber security analysts encounter.

To start this simulation, please click the "View Site" button below.

View Site

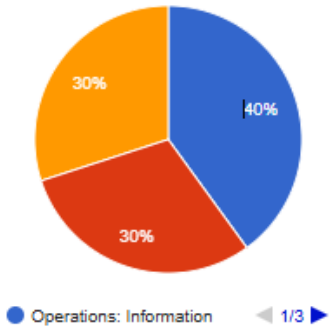
This action will open a "static site" on the right side of your screen. Follow the step-by-step instructions provided within the simulation to navigate through the events and locate the "flag." A flag is a series of characters with a format like this: "THM{RANDOM_WORDS}". Use this flag to answer questions from lessons here in TryHackMe, like the one below.

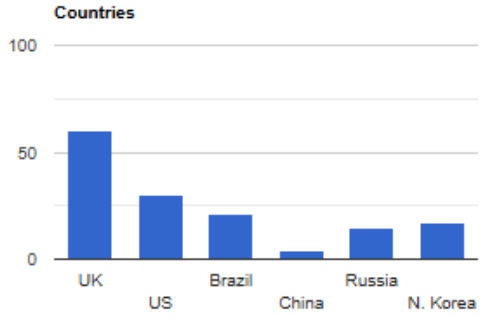
What's next?

In this lesson, we've discussed the different subfields (SOC, Threat Intelligence, Malware Analysis, and DFIR) and experienced firsthand how to deal with alerts in a simulated SIEM environment. While we've covered a lot, the depth and complexity of this field mean there's more to learn and explore. The lessons learned here will serve as your foundation as cyber threats evolve, demanding continuous learning, vigilance, and adaptation.

Continue learning by checking out the next lesson in this series, "Search Skills." This lesson will teach you valuable techniques for searching for information online to aid your investigations and learning.

https://siem.internal





Alert Log

Date	Message
Apr 5th 2026, 06:33:15:136	Successful SSH authentication attempt to port 22 from IP address 143.110.250.149
Apr 5th 2026, 06:30:30:144	Unauthorized connection attempt detected from IP address 143.110.250.149 to port 22
Apr 5th 2026, 04:13:01:244	The user John Doe logged in successfully (Event ID 4624)
Apr 5th 2026, 04:13:59:261	Multiple failed login attempts from John Doe
Apr 5th 2026, 04:03:57:254	Logon Failure: Specified Account's Password Has Expired (Event ID 535)

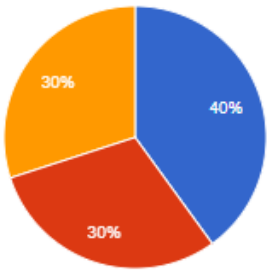
Suspicious red one

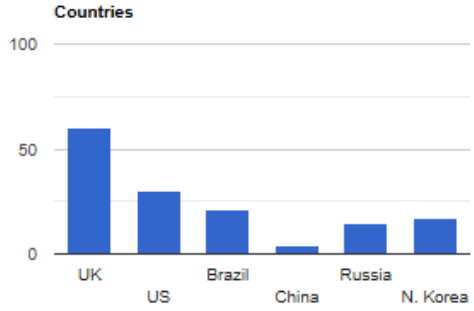
A Day In the Life of a Junior (Associate) Security Analyst

Instructions

Inspect the alerts in your SIEM dashboard. Find the malicious IP address from the alerts, make a note of it, and then click on the alert to proceed.

https://siem.internal





Operations: Information 1/3

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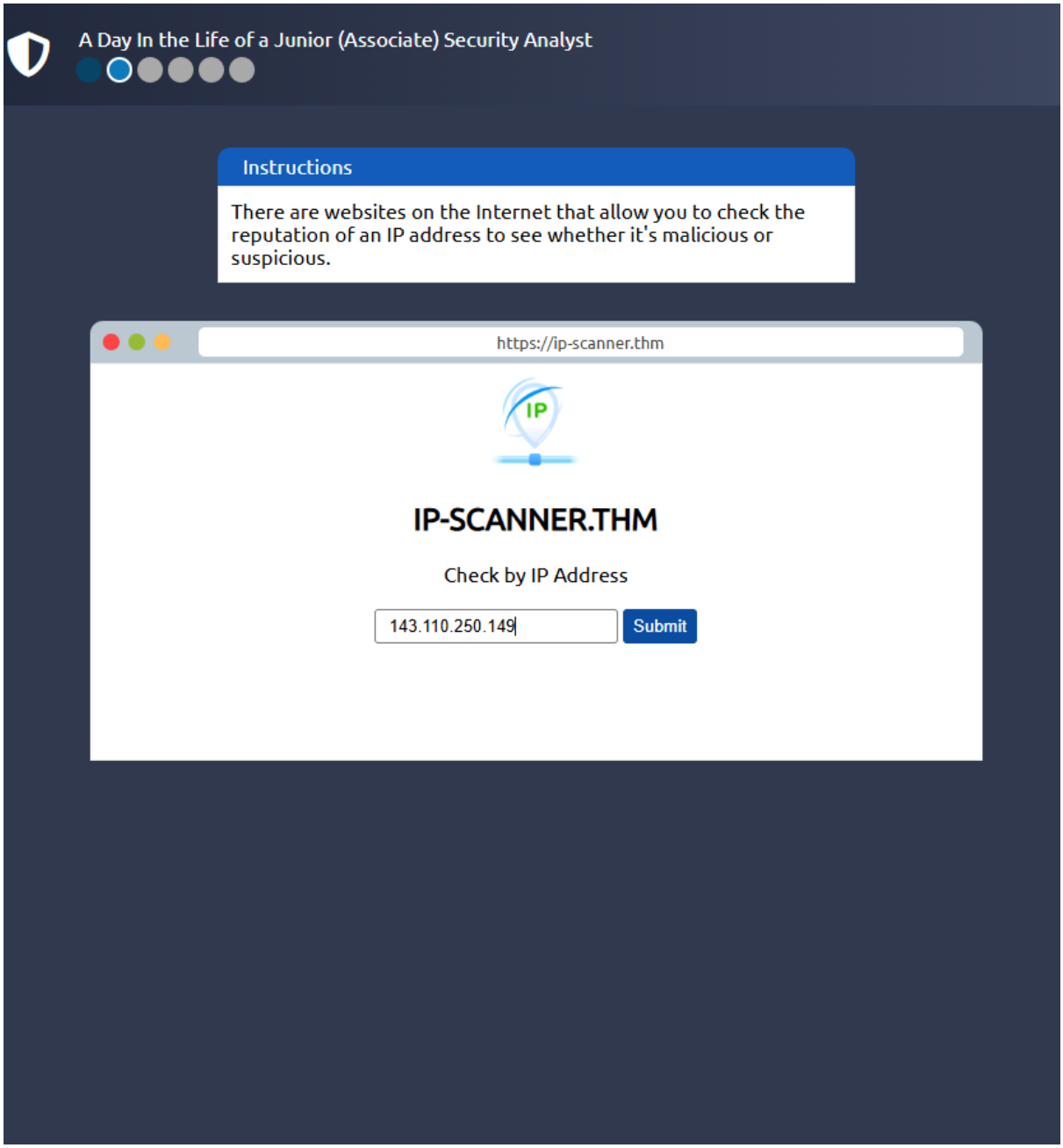
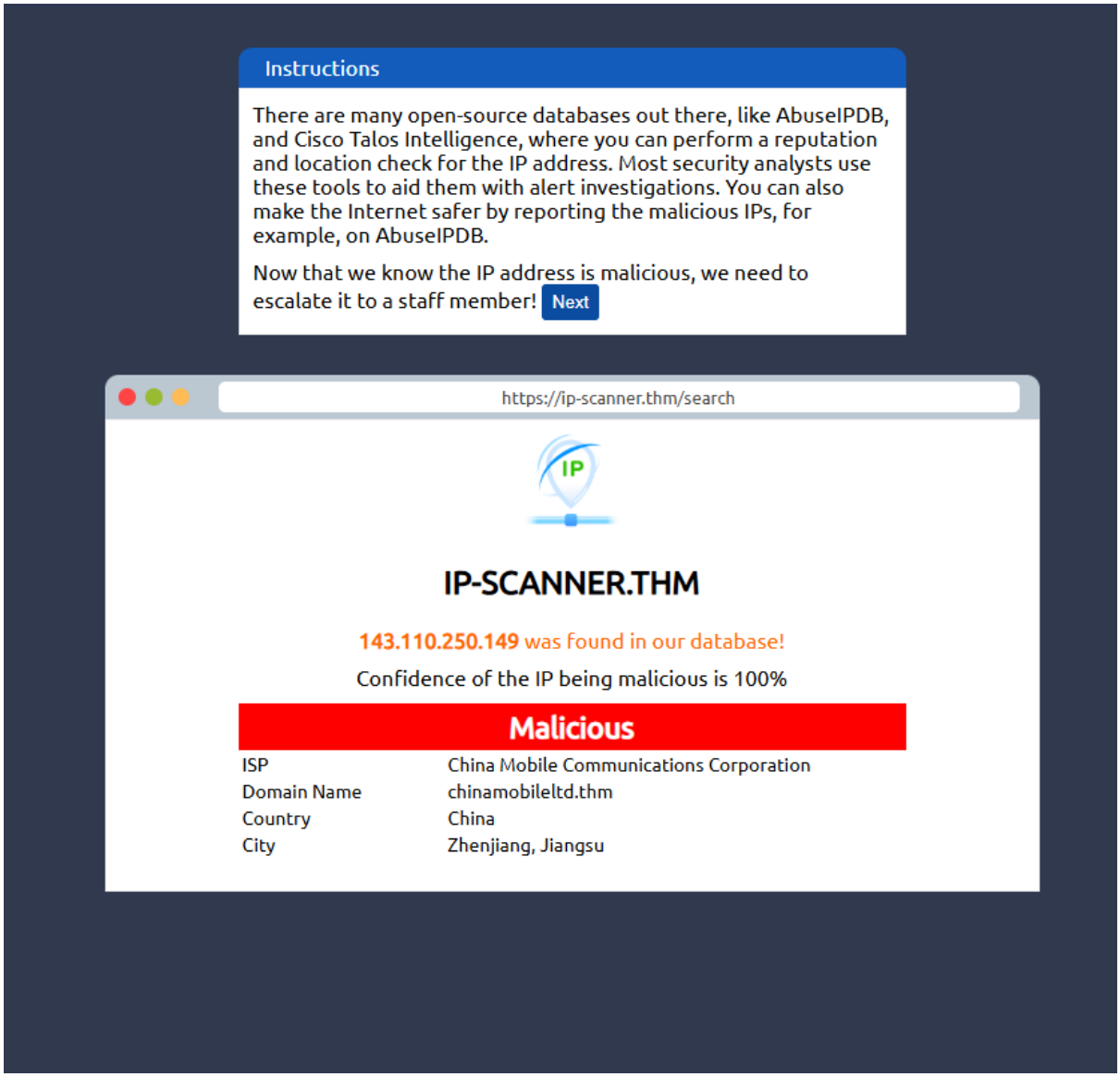
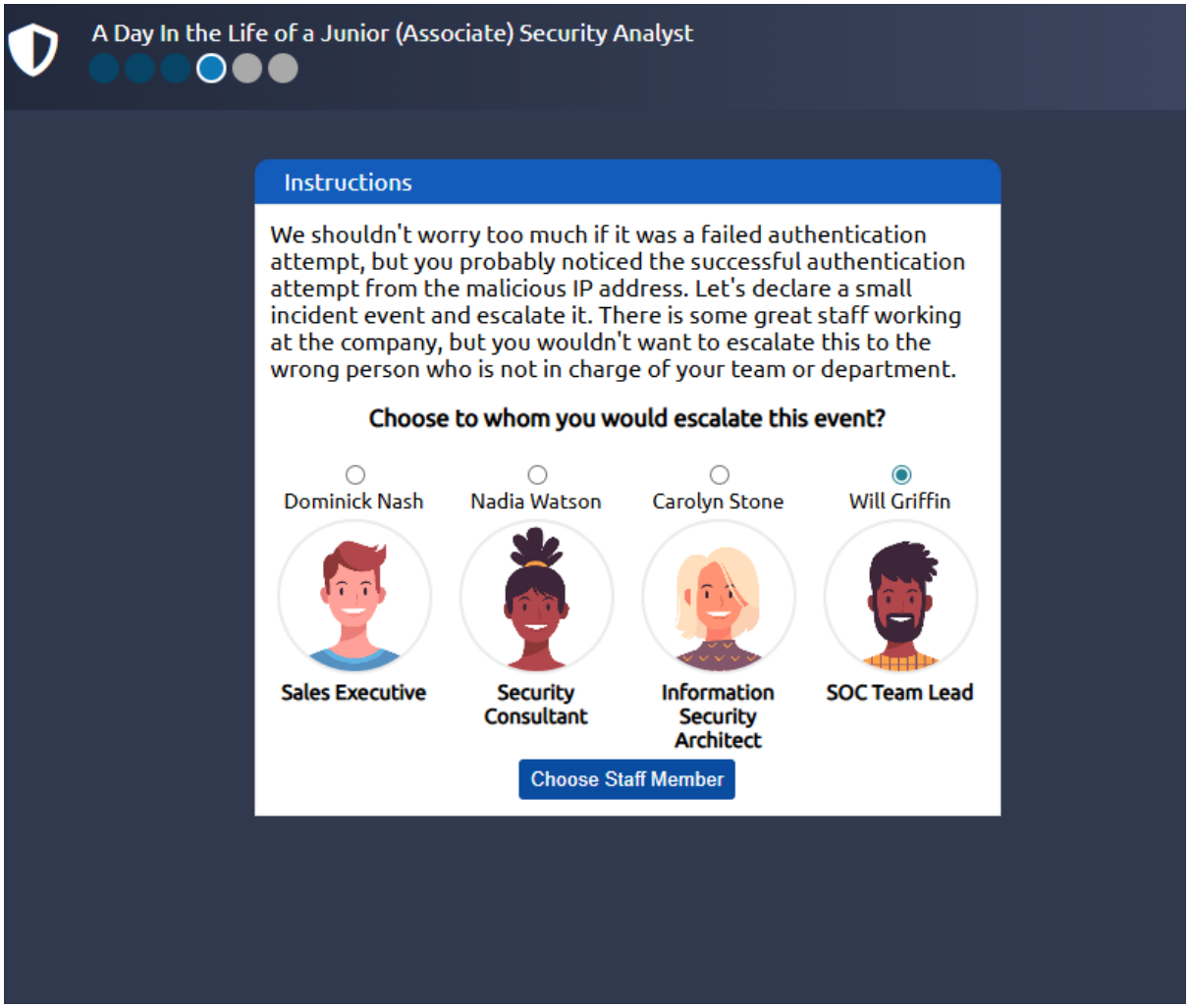
Click enter on red one and copy the Ip address



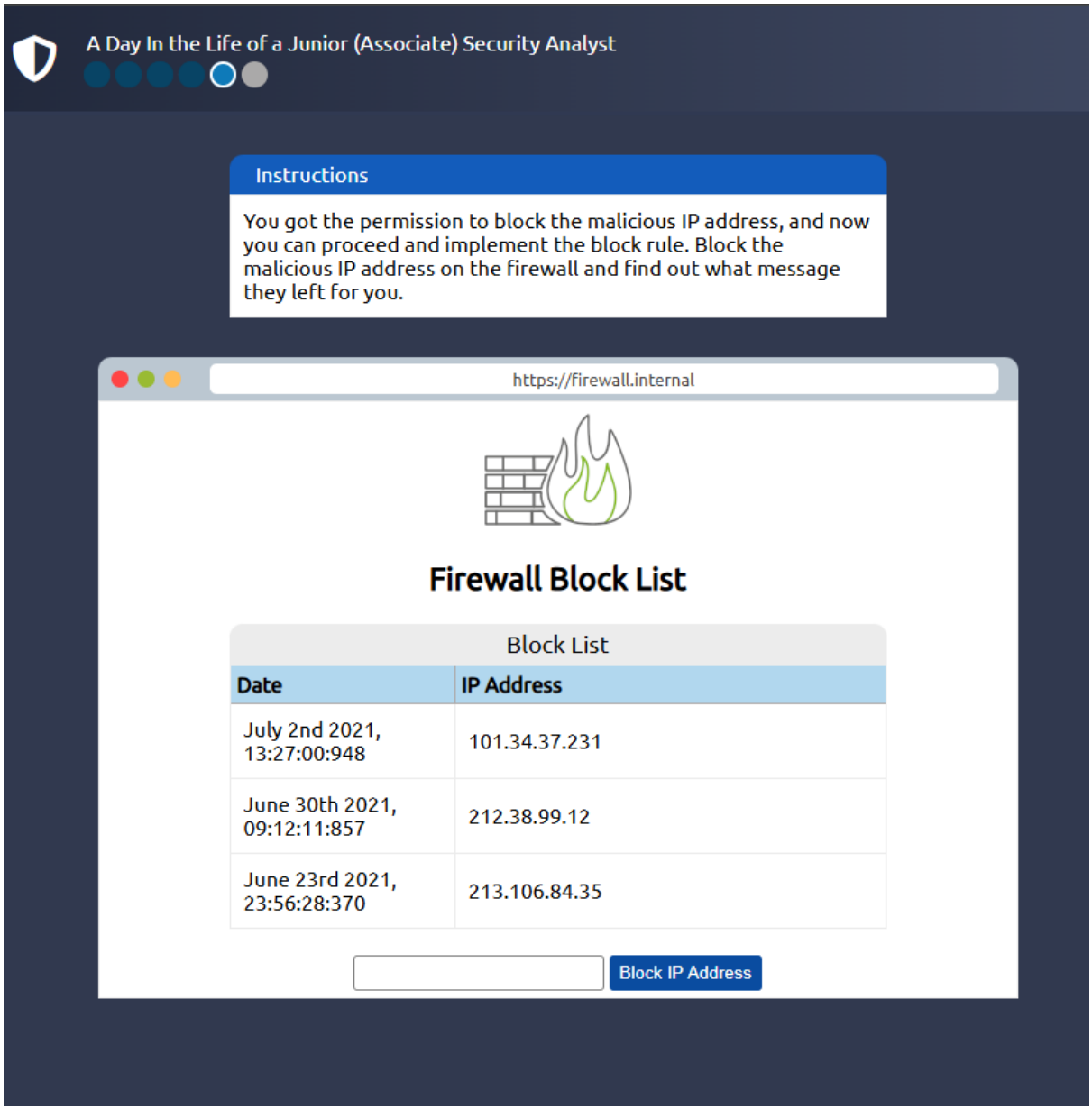
- Once you hit submit, you'll get the result, confirming that the suspected IP is "malicious".



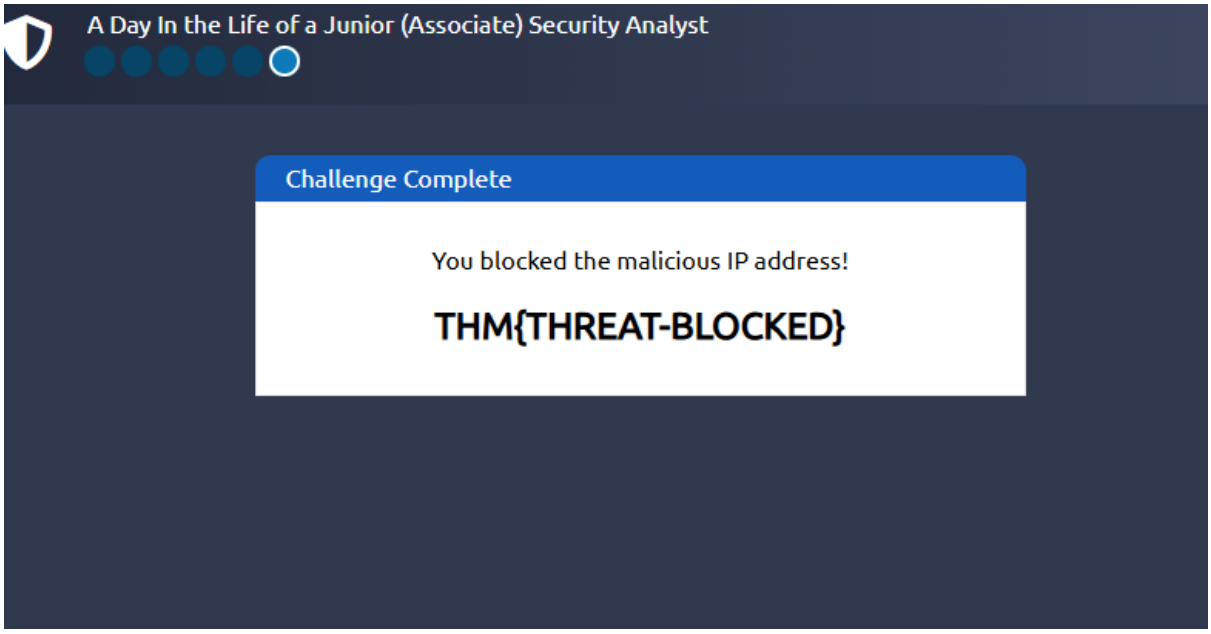
- As you have identified a security attack, it is the right time to escalate the event towards your team lead. You would be escalating the event to “SOC Team Lead” as you are a part of SOC team.



- Once he allows you to block the malicious IP address you just found, type in the malicious IP address in Firewall Block List and hit “Block IP Address”.



- Once the malicious IP address **143.110.250.149** is blocked, you'll be flashed your much awaiting flag i.e. **THM{THREAT-BLOCKED}.**



Answer the questions below

Q)What is the flag that you obtained by following along?

.A)**THM{THREAT-BLOCKED}.**

Conclusion

Today, we were introduced to defensive security and some of its roles. We also experienced a SOC analyst's job.